
Generalization in DQN with Actor

ActorDQN vs DQN

Xiaojie Gu

University of Wuerzburg

First Advisor: Mr. Sebastian Griesbach | Supervisor: Prof. Dr. Carlo D'Eramo

This presentation was made with assistance from GPT-6 Sol under the guidance of my hand draft.

Outline

1. Introduction
2. Environment design
3. Model Design
4. Experimental results

Introduction: generalization in robotics

Training condition



Table height: 1.0 m Garment: T-shirt
Neat fold in a familiar setting

Unseen test condition



Table height: 1.2 m Garment: jacket
Unsuccessful fold after the change

Success in training does not guarantee success when conditions change.

Images generated by GPT.

Generalization in robotics: Unitree video

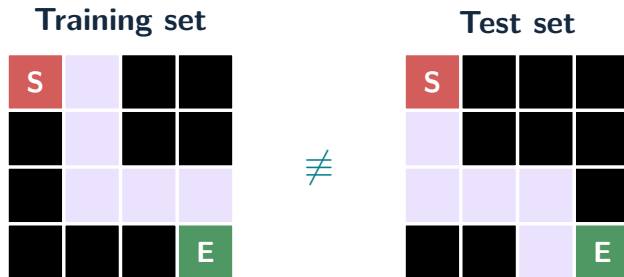


A real-world robotics example

Environment Design

1. Regulation
2. Procedural maze

Regulation: no shared optimal solution

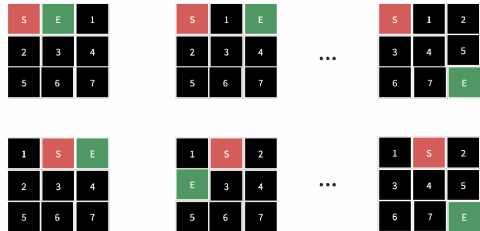


$$\mathcal{P}_{\text{train}} \cap \mathcal{P}_{\text{test}} = \emptyset$$

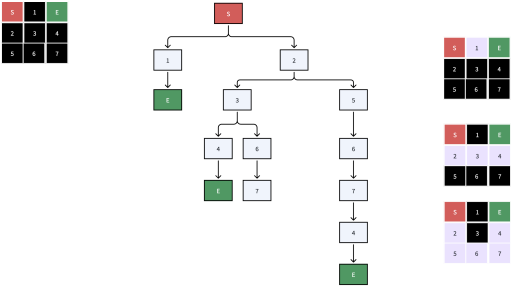
Each original maze and its variations stay in one split; the intended shortest path does not cross splits.

Procedural Maze

1. Enumerate start-goal pairs



2. Generate valid paths



Model Design

1. Idea: classification may be easier than regression
2. DQN
3. ActorDQN

Model Design: Why classification?

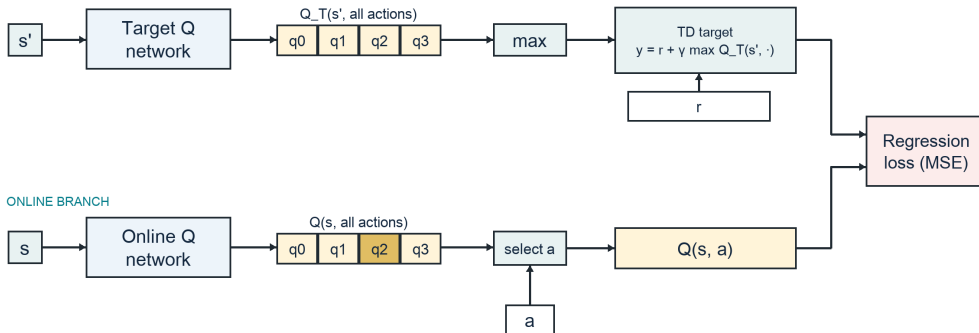
Hypothesis: Classification is easier. It might use a general pattern that leads to better generalization.

	Book 1	Book 2	Book 3	Book 4	
	0.1	0.2	0.4	0.9	1. score all four
Regression (two steps)	×	×	×	✓	2. choose the best
Classification	×	×	×	✓	learn the best-book label directly

Model Design: DQN

REPLAY TRANSITION (s, a, r, s')

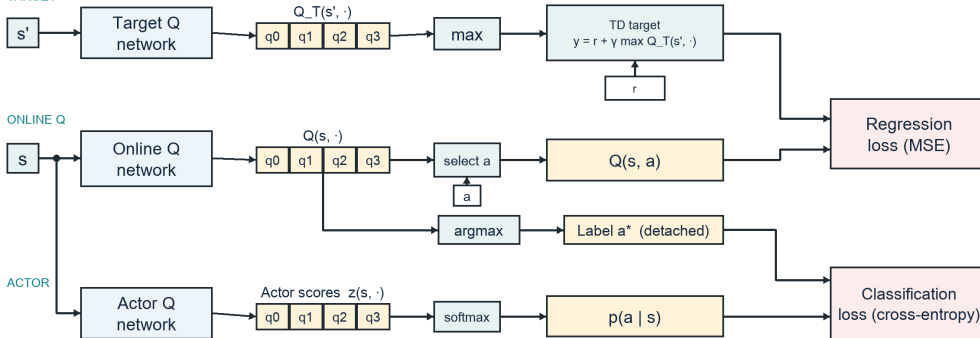
TARGET BRANCH



Model Design: ActorDQN

REPLAY TRANSITION (s, a, r, s')

TARGET



Experimental Results

1. Performance in mazes

Full held-out sets for 4×4 and 5×5 mazes.

2. Justification for the idea

Performance in Maze: 4×4

Training set: 803 | Evaluation set: 201
Step limit: 160

Policy	Success rate	Mean step	Optimal length
ActorDQN	0.828 ± 0.029	49.01 ± 6.60	6.163 ± 0.143
Basic DQN	0.111 ± 0.072	142.76 ± 11.22	6.163 ± 0.143
Uniform Random	0.740 ± 0.048	88.87 ± 5.74	6.163 ± 0.143

Performance in Maze: 5×5

Training set: 11,891 | Evaluation set: 2,973
Step limit: 250

Policy	Success rate	Mean step	Optimal length
ActorDQN	0.673 ± 0.027	114.11 ± 8.80	10.236 ± 0.014
Basic DQN	0.129 ± 0.036	218.71 ± 8.52	10.236 ± 0.014
Uniform Random	0.564 ± 0.004	178.01 ± 0.43	10.236 ± 0.014

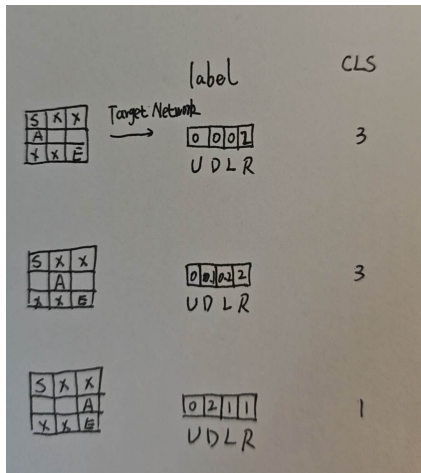
Generalization Gap

Gap = training-set mean – held-out mean;

Maze	Policy	Δ success rate	Δ mean step
4×4	ActorDQN	+0.026	−6.07
	Basic DQN	+0.102	−15.72
5×5	ActorDQN	+0.014	−3.89
	Basic DQN	+0.008	−1.83

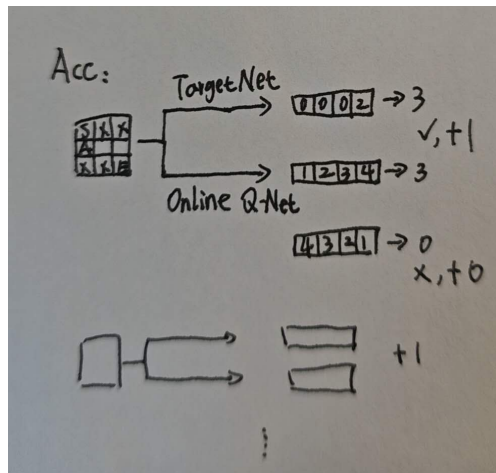
A smaller absolute gap is better.

Justification for the Idea



Classification vs. Q-value Regression

Target Network coming from	Task Type	here ↓ Train Eval □ □	
		Acc	Success Rate
DQN	CLS	<u>0.899</u>	137/201
	Reg	0.853	116/201
ActorDQN	CLS	<u>0.967</u>	175/201
	Reg	0.955	162/201



Thanks